Introduction to Database project

AIRLINE RESERVATION SYSTEM

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Students and Implemented SQL Queries

Student Name	SQL Queries Implemented
Duha	**Basic SELECT:** Passengers from Saudi Arabia, Flights going to Jeddah.**Aggregate Functions:** Count flights per airline, Average airplane capacity per airline.**Joins:** Bookings with passenger names, Flights with status information.**Subqueries:** Passengers who booked First-Class tickets, Flights operated by UAE airlines.**View:** Flight report view.
Danah	**Basic SELECT:** Male passengers information, Business class bookings.**Aggregate Functions:** Total number of airplanes, Bookings per flight.**Joins:** Flight departure time and status, Bookings with flight destinations.**Subqueries:** Scheduled flights, Flights using airplanes with capacity greater than 300.**View:** Airplane capacity view.
Wasan	**Basic SELECT:** Airbus models (A320, A380), Passengers born before 1990.**Aggregate Functions:** Total seating capacity per airline, Passenger count by nationality.**Joins:** Bookings linked to airlines, Passenger names with flight destinations.**Subqueries:** Flights operated by Saudi Airlines, Earliest departing flight, Passengers traveling from Riyadh, Airplanes above average capacity.**View:** High-capacity fleet report.
Retal	**Basic SELECT:** Passengers from Egypt, Flights departing from Riyadh.**Aggregate Functions:** Number of passengers per nationality, Average airplane capacity.**Joins:** Booking details with passenger names, Flights with airline names.**Subqueries:** Passengers who booked flights to Dubai, Flights using airplanes.

1. Conceptual and Logical Design

1.1 ER Model (Scenario):

✈ **Practical Scenario for Airline Database**

- The airline database includes multiple Airlines, each operating several Flights.

  - Each Airline has a unique Airline_ID, an Airline_Name, and a Country of registration.
  - Each airline operates several flights.

  - Each Airplane has a unique Airplane_ID, an Airplane_Number, a Model, and a Capacity.
  - Each airplane is assigned to exactly one flight, and each flight is assigned to exactly one airplane.

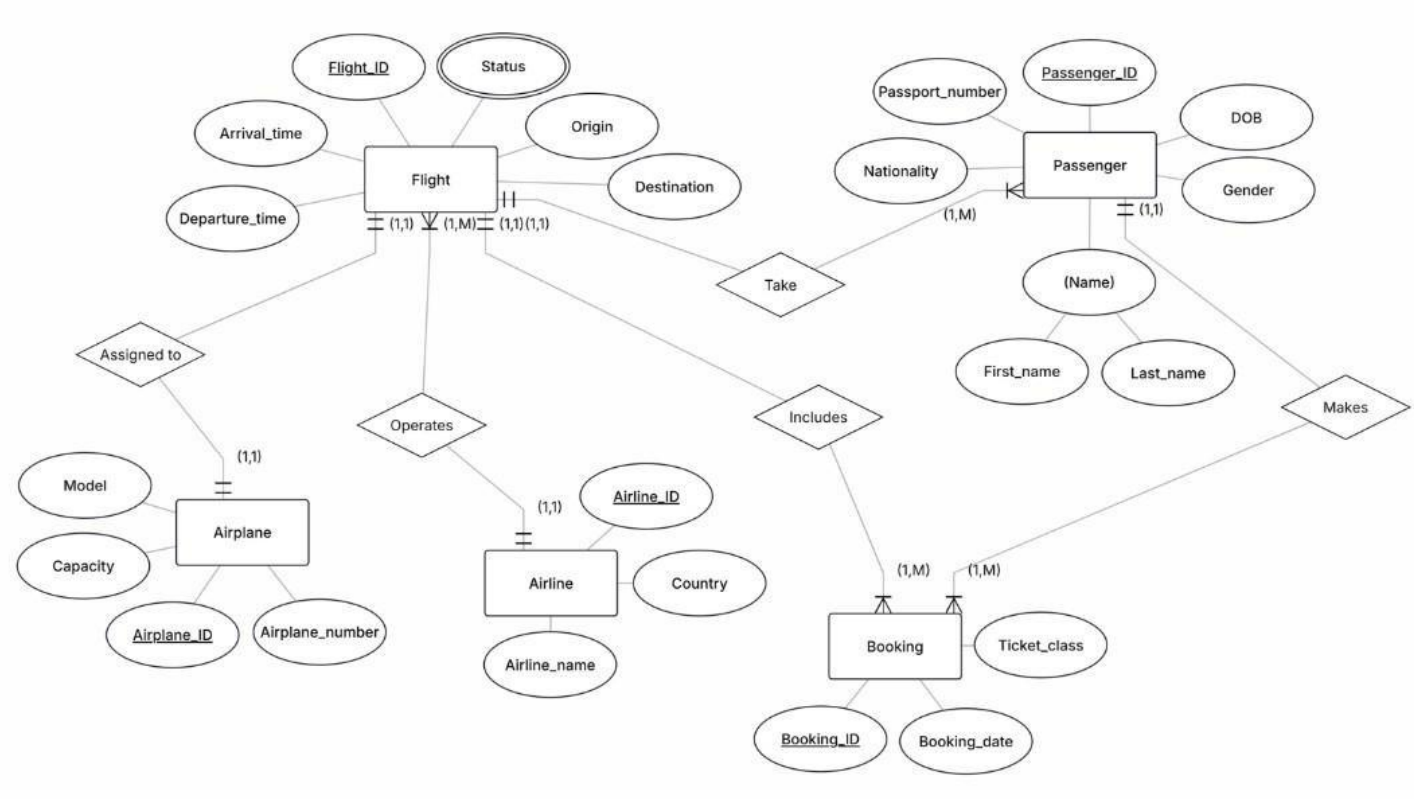
  - Each Flight has a unique Flight_ID, an Origin, a Destination, a Departure_Time, and an Arrival_Time. Each flight also has a Status (Multivalued) that indicates its current state: {Scheduled, Delayed, Departed, Arrived, Cancelled, Boarding}.
  - A flight is operated by one airline and assigned to one airplane.
  - A flight can include multiple bookings.

  - Each Booking has a unique Booking_ID, a Booking_Date, and a Ticket_Class.
  - A booking is included to one specific flight. And made by exactly one Passenger.

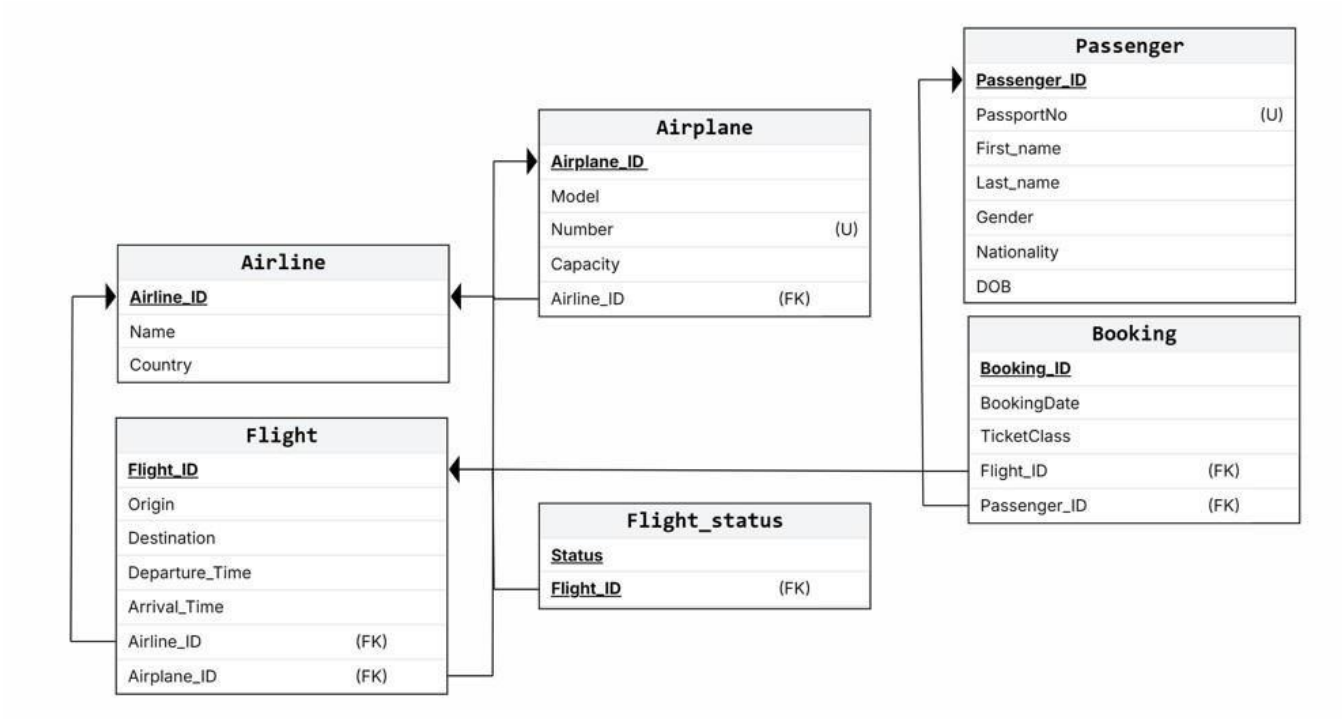
  - Each Passenger has a unique Passenger_ID, a First_Name, a Last_Name, a Gender, a Date_of_Birth, a Passport_Number, and a Nationality.

  - A passenger can make multiple bookings for different flights.

1.2 Entity Relationship Diagram (ERD):



1.3 Relational Schema:



2. Implementation:

2.1 Create Database

CREATE DATABASE airline_system: Creates a new database container named "airline_system"

USE airline_system : Sets the newly created database as the active workspace for all subsequent commands.

```
CREATE DATABASE airline_system;
USE airline_system;
```

2.1 Create tables:

1. CREATE TABLE Airline :

Airline_ID INT AUTO_INCREMENT PRIMARY KEY: Defines a unique identifier for each airline that increases automatically.

Name VARCHAR(100) NOT NULL: Stores the airline's name; it cannot be left empty.

Country VARCHAR(100) NOT NULL: Stores the country where the airline is based; it cannot be empty.

```
---------------------------------------
-- TABLE: Airline
---------------------------------------
CREATE TABLE Airline (
    Airline_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Name            VARCHAR(100) NOT NULL,
    Country         VARCHAR(100) NOT NULL
);

---------------------------------------
```

- Describe:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Airline_ID	int	NO	PRI	NULL	auto_increment
Name	varchar(100)	NO		NULL	
Country	varchar(100)	NO		NULL	

2. CREATE TABLE Passenger :

Passenger_ID INT AUTO_INCREMENT PRIMARY KEY,: A unique ID for each passenger.
PassportNo VARCHAR(20) NOT NULL UNIQUE,: Stores the passport number; it must be unique to prevent duplicate identities.
First_name VARCHAR(50) NOT NULL,: Stores the passenger's first name.
Last_name VARCHAR(50) NOT NULL,: Stores the passenger's last name.
Gender VARCHAR(10),: Stores the gender of the passenger.
Nationality VARCHAR(50),: Stores the passenger's country of citizenship.
DOB DATE,: Stores the passenger's Date of Birth.
Phone_Number VARCHAR(20),: Stores contact information.
CONSTRAINT chk_gender CHECK (Gender IN ('Male','Female')): Validation Rule:
Ensures that only 'Male' or 'Female' can be entered into the system.

```
----------------------------------------
-- TABLE: Passenger
----------------------------------------
CREATE TABLE Passenger (
    Passenger_ID    INT AUTO_INCREMENT PRIMARY KEY,
    PassportNo      VARCHAR(20) NOT NULL UNIQUE,
    First_name      VARCHAR(50) NOT NULL,
    Last_name       VARCHAR(50) NOT NULL,
    Gender          VARCHAR(10),
    Nationality     VARCHAR(50),
    DOB             DATE,
    Phone_Number    VARCHAR(20),

    CONSTRAINT chk_gender CHECK (Gender IN ('Male','Female'))
);
```

- Describe:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Passenger_ID	int	NO	PRI	NULL	auto_increment
PassportNo	varchar(20)	NO	UNI	NULL	
First_name	varchar(50)	NO		NULL	
Last_name	varchar(50)	NO		NULL	
Gender	varchar(10)	YES		NULL	
Nationality	varchar(50)	YES		NULL	
DOB	date	YES		NULL	
Phone_Number	varchar(20)	YES		NULL	

3. CREATE TABLE Airplane :

Airplane_ID INT AUTO_INCREMENT PRIMARY KEY,: A unique ID for each physical aircraft that increases automatically.

Model VARCHAR(50) NOT NULL,: Stores the specific aircraft model (e.g., Airbus A320).

Number VARCHAR(20) NOT NULL UNIQUE,: Stores the aircraft's tail number; it must be unique across the fleet.

Capacity INT NOT NULL,: Stores the maximum number of seats available for passengers. Airline_ID INT NOT NULL,: A foreign key field that creates a logical link between the plane and its owner (Airline).

CONSTRAINT chk_capacity CHECK (Capacity > 0),: Validation Rule: A integrity constraint that prevents entering zero or negative numbers for seating capacity.

FOREIGN KEY (Airline_ID) REFERENCES Airline (Airline_ID): Establishes a formal relational link to the Airline table.

ON DELETE RESTRICT: Referential Integrity: Prevents the deletion of an airline from the system if it still has airplanes registered to it.

ON UPDATE CASCADE: Referential Integrity: Automatically updates the Airline ID in this table if the ID is changed in the primary Airline table.

```
---------------------------------------
-- TABLE: Airplane
---------------------------------------
CREATE TABLE Airplane (
    Airplane_ID     INT AUTO_INCREMENT PRIMARY KEY,
    Model           VARCHAR(50) NOT NULL,
    Number          VARCHAR(20) NOT NULL UNIQUE,
    Capacity        INT NOT NULL,
    Airline_ID      INT NOT NULL,

    CONSTRAINT chk_capacity CHECK (Capacity > 0),

    FOREIGN KEY (Airline_ID)
        REFERENCES Airline (Airline_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);
```

- Describe:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Airplane_ID	int	NO	PRI	NULL	auto_increment
Model	varchar(50)	NO		NULL	
Number	varchar(20)	NO	UNI	NULL	
Capacity	int	NO		NULL	
Airline_ID	int	NO	MUL	NULL	

4. CREATE TABLE Flight :

Flight_ID INT AUTO_INCREMENT PRIMARY KEY,: A unique ID for each scheduled flight.

Origin VARCHAR(100) NOT NULL,: Stores the departure city/airport.

Destination VARCHAR(100) NOT NULL,: Stores the arrival city/airport.

Departure_Time DATETIME NOT NULL,: Stores the exact date and time of departure.

Arrival_Time DATETIME NOT NULL,: Stores the estimated arrival date and time.

Airline_ID INT NOT NULL,: Links the flight to the airline operating it.

Airplane_ID INT NOT NULL,: Links the flight to the specific aircraft being used.

FOREIGN KEY (Airplane_ID) REFERENCES Airplane (Airplane_ID): Ensures the plane used for the flight exists in the fleet.

```
-----------------------------------------
-- TABLE: Flight
-----------------------------------------
CREATE TABLE Flight (
    Flight_ID       INT AUTO_INCREMENT PRIMARY KEY,
    Origin          VARCHAR(100) NOT NULL,
    Destination     VARCHAR(100) NOT NULL,
    Departure_Time  DATETIME NOT NULL,
    Arrival_Time    DATETIME NOT NULL,
    Airline_ID      INT NOT NULL,
    Airplane_ID     INT NOT NULL,

    FOREIGN KEY (Airline_ID)
        REFERENCES Airline (Airline_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    FOREIGN KEY (Airplane_ID)
        REFERENCES Airplane (Airplane_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);
```

- Describe:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	auto_increment
Origin	varchar(100)	NO		NULL	
Destination	varchar(100)	NO		NULL	
Departure_Time	datetime	NO		NULL	
Arrival_Time	datetime	NO		NULL	
Airline_ID	int	NO	MUL	NULL	
Airplane_ID	int	NO	MUL	NULL	

5. CREATE TABLE Flight_Status :

Flight_ID INT NOT NULL,: Links the status to a specific flight.

Status VARCHAR(30) NOT NULL,: Stores the current state (e.g., Delayed, Boarding).

PRIMARY KEY (Flight_ID, Status),: Creates a composite primary key, allowing a flight to have multiple statuses over time (history tracking).

CONSTRAINT chk_status CHECK (Status IN (...)): Business Rule: Limits status entries to specific valid categories (Scheduled, Delayed, etc.).

ON DELETE CASCADE: Automatically removes the status history if the flight record is deleted.

```
-----------------------------------------
-- TABLE: Flight_Status MV
-----------------------------------------
CREATE TABLE Flight_Status (
    Flight_ID   INT NOT NULL,
    Status      VARCHAR(30) NOT NULL,
    PRIMARY KEY (Flight_ID, Status),

    CONSTRAINT chk_status CHECK (Status IN
        ('Scheduled','Delayed','Cancelled','Departed','Arrived', 'Boarding')
    ),

    FOREIGN KEY (Flight_ID)
        REFERENCES Flight (Flight_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

- Describe:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Status	varchar(30)	NO	PRI	NULL	

6. CREATE TABLE Booking :

Booking_ID INT AUTO_INCREMENT PRIMARY KEY,: A unique ID for each ticket reservation.

BookingDate DATETIME NOT NULL,: Stores when the ticket was purchased. TicketClass VARCHAR(20) NOT NULL,: Stores the service level (Economy, Business, First).

Flight_ID INT NOT NULL,: Links the booking to a specific flight.

Passenger_ID INT NOT NULL,: Links the booking to the passenger who purchased it.

CONSTRAINT chk_class CHECK (TicketClass IN ('Economy','Business','First')):

Business Rule: Restricts the booking to these three specific ticket classes.

FOREIGN KEY (Passenger_ID) REFERENCES Passenger (Passenger_ID): Ensures the passenger exists in the database.

```sql
-----------------------------------------
-- TABLE: Booking
-----------------------------------------
CREATE TABLE Booking (
    Booking_ID      INT AUTO_INCREMENT PRIMARY KEY,
    BookingDate     DATETIME NOT NULL,
    TicketClass     VARCHAR(20) NOT NULL,
    Flight_ID       INT NOT NULL,
    Passenger_ID    INT NOT NULL,

    CONSTRAINT chk_class CHECK (TicketClass IN ('Economy','Business','First')),

    FOREIGN KEY (Flight_ID)
        REFERENCES Flight (Flight_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    FOREIGN KEY (Passenger_ID)
        REFERENCES Passenger (Passenger_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);
```

- Describe:

Field	Type	Null	Key	Default	Extra
Booking_ID	int	NO	PRI	NULL	auto_increment
BookingDate	datetime	NO		NULL	
TicketClass	varchar(20)	NO		NULL	
Flight_ID	int	NO	MUL	NULL	
Passenger_ID	int	NO	MUL	NULL	

2.3 Inset Records

- Airline

```
INSERT INTO Airline (Name, Country) VALUES
('Saudi Airlines','Saudi Arabia'),
('Flynas','Saudi Arabia'),
('Qatar Airways','Qatar'),
('Emirates','UAE'),
('Etihad Airways','UAE'),
('Turkish Airlines','Turkey'),
('Lufthansa','Germany'),
('British Airways','UK'),
('EgyptAir','Egypt'),
('Oman Air','Oman');
```

Airline_ID	Name	Country
1	Saudi Airlines	Saudi Arabia
2	Flynas	Saudi Arabia
3	Qatar Airways	Qatar
4	Emirates	UAE
5	Etihad Airways	UAE
6	Turkish Airlines	Turkey
7	Lufthansa	Germany
8	British Airways	UK
9	EgyptAir	Egypt
10	Oman Air	Oman
NULL	NULL	NULL

- Passenger

```
INSERT INTO Passenger (PassportNo, First_name, Last_name, Gender, Nationality, DOB, Phone_Number) VALUES
('P1001','Ali','Saleh','Male','Saudi Arabia','1990-05-12','0551234567'),
('P1002','Sara','Ahmed','Female','Egypt','1995-02-20','0102233445'),
('P1003','Fahad','Harbi','Male','Saudi Arabia','1988-11-03','0508887766'),
('P1004','Maryam','Saad','Female','UAE','1992-01-15','0521113344'),
('P1005','John','Smith','Male','UK','1985-07-09','0044123456'),
('P1006','Lina','Hassan','Female','Jordan','1998-03-22','0795566778'),
('P1007','Omar','Farouq','Male','Egypt','1999-08-18','0124455667'),
('P1008','Nora','Ali','Female','Saudi Arabia','2000-12-01','0556677889'),
('P1009','Huda','Salem','Female','Qatar','1993-09-14','009743334455'),
('P1010','James','Brown','Male','USA','1990-10-05','0012023344');
```

Passenger_ID	PassportNo	First_name	Last_name	Gender	Nationality	DOB	Phone_Number
1	P1001	Ali	Saleh	Male	Saudi Arabia	1990-05-12	0551234567
2	P1002	Sara	Ahmed	Female	Egypt	1995-02-20	0102233445
3	P1003	Fahad	Harbi	Male	Saudi Arabia	1988-11-03	0508887766
4	P1004	Maryam	Saad	Female	UAE	1992-01-15	0521113344
5	P1005	John	Smith	Male	UK	1985-07-09	0044123456
6	P1006	Lina	Hassan	Female	Jordan	1998-03-22	0795566778
7	P1007	Omar	Farouq	Male	Egypt	1999-08-18	0124455667
8	P1008	Nora	Ali	Female	Saudi Arabia	2000-12-01	0556677889
9	P1009	Huda	Salem	Female	Qatar	1993-09-14	009743334455
10	P1010	James	Brown	Male	USA	1990-10-05	0012023344
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Airplane

```
INSERT INTO Airplane (Model, Number, Capacity, Airline_ID) VALUES
('A320','SA100',180,1),
('A330','SA200',250,1),
('B777','FN300',300,2),
('A350','QA400',320,3),
('B787','EM500',250,4),
('B737','ET600',160,5),
('A380','EM700',500,4),
('B747','TK800',410,6),
('A321','LH900',190,7),
('B767','BA1000',200,8);
```

Airplane_ID	Model	Number	Capacity	Airline_ID
1	A320	SA100	180	1
2	A330	SA200	250	1
3	B777	FN300	300	2
4	A350	QA400	320	3
5	B787	EM500	250	4
6	B737	ET600	160	5
7	A380	EM700	500	4
8	B747	TK800	410	6
9	A321	LH900	190	7
10	B767	BA1000	200	8
NULL	NULL	NULL	NULL	NULL

- Flight

```
INSERT INTO Flight (Origin, Destination, Departure_Time, Arrival_Time, Airline_ID, Airplane_ID) VALUES
('Riyadh','Jeddah','2025-01-01 08:00','2025-01-01 10:00',1,1),
('Jeddah','Dubai','2025-01-02 12:00','2025-01-02 15:00',1,2),
('Doha','London','2025-01-03 05:00','2025-01-03 12:30',3,4),
('Dubai','Istanbul','2025-01-04 09:00','2025-01-04 13:00',4,5),
('Istanbul','Frankfurt','2025-01-05 14:00','2025-01-05 17:00',6,8),
('Riyadh','Cairo','2025-01-06 07:00','2025-01-06 09:30',2,3),
('Jeddah','Doha','2025-01-07 11:00','2025-01-07 13:00',1,1),
('London','Dubai','2025-01-08 21:00','2025-01-09 05:00',8,10),
('Cairo','Jeddah','2025-01-09 06:00','2025-01-09 08:00',9,9),
('Jeddah','Muscat','2025-01-10 16:00','2025-01-10 18:00',10,10);
```

Flight_ID	Origin	Destination	Departure_Time	Arrival_Time	Airline_ID	Airplane_ID
1	Riyadh	Jeddah	2025-01-01 08:00:00	2025-01-01 10:00:00	1	1
2	Jeddah	Dubai	2025-01-02 12:00:00	2025-01-02 15:00:00	1	2
3	Doha	London	2025-01-03 05:00:00	2025-01-03 12:30:00	3	4
4	Dubai	Istanbul	2025-01-04 09:00:00	2025-01-04 13:00:00	4	5
5	Istanbul	Frankfurt	2025-01-05 14:00:00	2025-01-05 17:00:00	6	8
6	Riyadh	Cairo	2025-01-06 07:00:00	2025-01-06 09:30:00	2	3
7	Jeddah	Doha	2025-01-07 11:00:00	2025-01-07 13:00:00	1	1
8	London	Dubai	2025-01-08 21:00:00	2025-01-09 05:00:00	8	10
9	Cairo	Jeddah	2025-01-09 06:00:00	2025-01-09 08:00:00	9	9
10	Jeddah	Muscat	2025-01-10 16:00:00	2025-01-10 18:00:00	10	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Flight_Status

```
INSERT INTO Flight_Status VALUES
(1,'Scheduled'),
(2,'Scheduled'),
(2,'Delayed'),
(3,'Scheduled'),
(4,'Cancelled'),
(5,'Arrived'),
(6,'Scheduled'),
(6,'Departed'),
(7,'Scheduled'),
(8,'Scheduled'),
(9,'Arrived'),
(10,'Scheduled');
```

Flight_ID	Status
1	Scheduled
2	Delayed
2	Scheduled
3	Scheduled
4	Cancelled
5	Arrived
6	Departed
6	Scheduled
7	Scheduled
8	Scheduled
9	Arrived
10	Scheduled
NULL	NULL

- Booking

```
INSERT INTO Booking (BookingDate, TicketClass, Flight_ID, Passenger_ID) VALUES
('2025-01-01 07:00','Economy',1,1),
('2025-01-01 07:30','Business',1,2),
('2025-01-02 11:00','Economy',2,3),
('2025-01-03 03:00','First',3,4),
('2025-01-04 08:00','Economy',4,5),
('2025-01-05 10:00','Business',5,6),
('2025-01-06 05:30','Economy',6,7),
('2025-01-07 09:40','Business',7,8),
('2025-01-08 20:00','Economy',8,9),
('2025-01-09 05:15','First',9,10);
```

Booking_ID	BookingDate	TicketClass	Flight_ID	Passenger_ID
1	2025-01-01 07:00:00	Economy	1	1
2	2025-01-01 07:30:00	Business	1	2
3	2025-01-02 11:00:00	Economy	2	3
4	2025-01-03 03:00:00	First	3	4
5	2025-01-04 08:00:00	Economy	4	5
6	2025-01-05 10:00:00	Business	5	6
7	2025-01-06 05:30:00	Economy	6	7
8	2025-01-07 09:40:00	Business	7	8
9	2025-01-08 20:00:00	Economy	8	9
10	2025-01-09 05:15:00	First	9	10
NULL	NULL	NULL	NULL	NULL

2.4 Duha's Queries

1) Basic SELECT

Passengers from Saudi Arabia

This query retrieves the first name, last name, and nationality of all passengers from the Passenger table whose nationality is specifically 'Saudi Arabia'.

```
---------------------------------------
-- Duha's QUERIES
---------------------------------------
-- Show all passengers from Saudi Arabia
SELECT First_name, Last_name, Nationality
FROM Passenger
WHERE Nationality = 'Saudi Arabia';
```

Output:

Result Grid | Filter Rows:

First_name	Last_name	Nationality
Ali	Saleh	Saudi Arabia
Fahad	Harbi	Saudi Arabia
Nora	Ali	Saudi Arabia

Flights going to Jeddah

This query displays the flight ID, origin, destination, and departure time for all flights where the destination is 'Jeddah'.

```
-- Show all flights going to Jeddah
SELECT Flight_ID, Origin, Destination, Departure_Time
FROM Flight
WHERE Destination = 'Jeddah';
```

Output:

Result Grid | Filter Rows: | Edit:

Flight_ID	Origin	Destination	Departure_Time
1	Riyadh	Jeddah	2025-01-01 08:00:00
9	Cairo	Jeddah	2025-01-09 06:00:00
NULL	NULL	NULL	NULL

2) Aggregate Functions

Total flights per airline

This query counts how many flights each airline operates by joining the Airline and Flight tables and grouping the results by the airline's name.

```
-- Count how many flights each airline operates
SELECT Airline.Name AS Airline_Name, COUNT(Flight.Flight_ID) AS Total_Flights
FROM Airline
JOIN Flight ON Airline.Airline_ID = Flight.Airline_ID
GROUP BY Airline.Name;
```

Output:

Airline_Name	Total_Flights
Saudi Airlines	3
Flynas	1
Qatar Airways	1
Emirates	1
Turkish Airlines	1
British Airways	1
EgyptAir	1
Oman Air	1

Average airplane capacity per airline

This query calculates the average seating capacity for all airplanes belonging to each airline, grouping the final results by airline name.

```
-- Average airplane capacity per airline
SELECT Airline.Name AS Airline_Name, AVG(Airplane.Capacity) AS Avg_Capacity
FROM Airline
JOIN Airplane ON Airline.Airline_ID = Airplane.Airline_ID
GROUP BY Airline.Name;
```

Output:

Airline_Name	Avg_Capacity
Saudi Airlines	215.0000
Flynas	300.0000
Qatar Airways	320.0000
Emirates	375.0000
Etihad Airways	160.0000
Turkish Airlines	410.0000
Lufthansa	190.0000
British Airways	200.0000

3) Join Between Tables

Bookings with passenger names

This query joins the Booking and Passenger tables to display the booking ID alongside the first and last names of the passengers to identify who made each reservation.

```
-- Show bookings with passenger names
SELECT Booking.Booking_ID, Passenger.First_name, Passenger.Last_name, Booking.TicketClass
FROM Booking
INNER JOIN Passenger ON Booking.Passenger_ID = Passenger.Passenger_ID;
```

Output:

Result Grid | Filter Rows: | Export:

Booking_ID	First_name	Last_name	TicketClass
1	Ali	Saleh	Economy
2	Sara	Ahmed	Business
3	Fahad	Harbi	Economy
4	Maryam	Saad	First
5	John	Smith	Economy
6	Lina	Hassan	Business
7	Omar	Farouq	Economy
8	Nora	Ali	Business
9	Huda	Salem	Economy
10	James	Brown	First

Result 20

Flight status details

This query links the Flight table with the Flight_Status table to show operational details for each flight, such as whether a trip is scheduled, delayed, or arrived.

Limit to 1000 rows

```
-- Show flights and the airplane model used
SELECT Flight.Flight_ID, Flight.Origin, Flight.Destination, Flight_Status.Status
FROM Flight
LEFT JOIN Flight_Status
    ON Flight.Flight_ID = Flight_Status.Flight_ID;
```

Output:

Result Grid | Filter Rows:

Flight_ID	Origin	Destination	Status
1	Riyadh	Jeddah	Scheduled
2	Jeddah	Dubai	Delayed
2	Jeddah	Dubai	Scheduled
3	Doha	London	Scheduled
4	Dubai	Istanbul	Cancelled
5	Istanbul	Frankfurt	Arrived
6	Riyadh	Cairo	Departed
6	Riyadh	Cairo	Scheduled
7	Jeddah	Doha	Scheduled
8	London	Dubai	Scheduled

4) Subqueries to Answer Specific Questions

Passengers with First-Class tickets

This subquery identifies the names of passengers who have specifically booked tickets in the 'First' class category by filtering IDs found in the Booking table.

```
-- Passengers who booked a First-Class ticket
SELECT First_name, Last_name
FROM Passenger
WHERE Passenger_ID IN (
    SELECT Passenger_ID
    FROM Booking
    WHERE TicketClass = 'First'
);
```

Output:

Result Grid | Filter Rows:

First_name	Last_name
Maryam	Saad
James	Brown

Flights from UAE airlines

This subquery retrieves flight information for all companies whose country of operation is listed as 'UAE' within the Airline table.

```
-- Flights operated by airlines located in the UAE
SELECT Flight_ID, Origin, Destination
FROM Flight
WHERE Airline_ID IN (
    SELECT Airline_ID
    FROM Airline
    WHERE Country = 'UAE'
);
```

Output:

Result Grid | Filter Rows:

Flight_ID	Origin	Destination
4	Dubai	Istanbul
NULL	NULL	NULL

5) View

Flight Status Report View

This view creates a comprehensive summary report by joining the Flight and Flight_Status tables. It displays essential journey details including the flight ID, origin, destination, and exact departure and arrival times alongside the current real-time status of each trip. This report is designed to provide a quick operational overview of all active flights in the system.

```
-- Create a view showing all flights and their status
CREATE VIEW Flight_Report AS
SELECT Flight.Flight_ID, Flight.Origin, Flight.Destination,
       Flight.Departure_Time, Flight.Arrival_Time,
       Flight_Status.Status
FROM Flight
JOIN Flight_Status ON Flight.Flight_ID = Flight_Status.Flight_ID;
SELECT * FROM Flight_Report;
describe flight_report;
```

Output:

```
-- Create a view showing all flights and their status
CREATE VIEW Flight_Report AS
SELECT Flight.Flight_ID, Flight.Origin, Flight.Destination,
       Flight.Departure_Time, Flight.Arrival_Time,
       Flight_Status.Status
FROM Flight
JOIN Flight_Status ON Flight.Flight_ID = Flight_Status.Flight_ID;
SELECT * FROM Flight_Report;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Flight_ID	Origin	Destination	Departure_Time	Arrival_Time	Status
1	Riyadh	Jeddah	2025-01-01 08:00:00	2025-01-01 10:00:00	Scheduled
2	Jeddah	Dubai	2025-01-02 12:00:00	2025-01-02 15:00:00	Scheduled

2.5 Danah's QUERIES

1) Basic SELECT

Male passengers information

This query retrieves the first name, last name, and date of birth of all male passengers.

```
-- Show the First Name, Last Name, and Date of Birth of all male passengers
SELECT First_name, Last_name, DOB
FROM Passenger
WHERE Gender = 'Male';
```

Output:

First_name	Last_name	DOB
Ali	Saleh	1990-05-12
Fahad	Harbi	1988-11-03
John	Smith	1985-07-09
Omar	Farouq	1999-08-18
James	Brown	1990-10-05

Business class bookings

This query displays booking ID, booking date, and ticket class for bookings made in Business class.

```
-- Show the Booking ID, Booking Date, and Ticket Class for all bookings made
SELECT Booking_ID, BookingDate, TicketClass
FROM Booking
WHERE TicketClass = 'Business';
```

Output:

Booking_ID	BookingDate	TicketClass
2	2025-01-01 07:30:00	Business
6	2025-01-05 10:00:00	Business
8	2025-01-07 09:40:00	Business
NULL	NULL	NULL

2) Aggregate Functions

Total number of airplanes

This query counts the total number of airplanes in the database.

```
-- Count the total number of airplanes in the database
SELECT COUNT(Airplane_ID) AS Total_Airplanes_Count
FROM Airplane;
```

Output:

	Total_Airplanes_Count
▶	10

Bookings per flight

This query counts the number of bookings associated with each flight.

```
-- Count the number of bookings made for each flight
SELECT Flight_ID, COUNT(Booking_ID) AS Total_Bookings_Per_Flight
FROM Booking
GROUP BY Flight_ID;
```

Output:

	Flight_ID	Total_Bookings_Per_Flight
▶	1	2
	2	1
	3	1
	4	1
	5	1
	6	1
	7	1
	8	1
	9	1

3) Join Between Tables

Flight departure time and status

This query joins the Flight and Flight_Status tables to show each flight's departure time and current status.

```
-- List the Flight ID, Departure Time, and its recorded Status
SELECT F.Flight_ID, F.Departure_Time, FS.Status
FROM Flight F
INNER JOIN Flight_Status FS
    ON F.Flight_ID = FS.Flight_ID;
```

Output:

Flight_ID	Departure_Time	Status
1	2025-01-01 08:00:00	Scheduled
2	2025-01-02 12:00:00	Delayed
2	2025-01-02 12:00:00	Scheduled
3	2025-01-03 05:00:00	Scheduled
4	2025-01-04 09:00:00	Cancelled
5	2025-01-05 14:00:00	Arrived
6	2025-01-06 07:00:00	Departed
6	2025-01-06 07:00:00	Scheduled
7	2025-01-07 11:00:00	Scheduled
8	2025-01-08 21:00:00	Scheduled
9	2025-01-09 06:00:00	Arrived
10	2025-01-10 16:00:00	Scheduled

Bookings with flight destinations

This query joins the Booking and Flight tables to display bookings along with flight destinations.

```
-- Show all Bookings and the destination of the booked flight
SELECT B.Booking_ID, B.BookingDate, F.Destination AS Flight_Destination
FROM Booking B
LEFT JOIN Flight F
    ON B.Flight_ID = F.Flight_ID;
```

Output:

Booking_ID	BookingDate	Flight_Destination
1	2025-01-01 07:00:00	Jeddah
2	2025-01-01 07:30:00	Jeddah
3	2025-01-02 11:00:00	Dubai
4	2025-01-03 03:00:00	London
5	2025-01-04 08:00:00	Istanbul
6	2025-01-05 10:00:00	Frankfurt
7	2025-01-06 05:30:00	Cairo
8	2025-01-07 09:40:00	Doha
9	2025-01-08 20:00:00	Dubai
10	2025-01-09 05:15:00	Jeddah

4) Subqueries to Answer Specific Questions

Scheduled flights

This subquery retrieves flights that currently have a "Scheduled" status.

```
-- Show all flights that currently have a 'Scheduled' status
SELECT Flight_ID, Origin, Destination
FROM Flight
WHERE Flight_ID IN (
    SELECT Flight_ID
    FROM Flight_Status
    WHERE Status = 'Scheduled'
);
```

Output:

Flight_ID	Origin	Destination
1	Riyadh	Jeddah
2	Jeddah	Dubai
3	Doha	London
6	Riyadh	Cairo
7	Jeddah	Doha
8	London	Dubai
10	Jeddah	Muscat
NULL	NULL	NULL

Flights using airplanes with capacity greater than 300 This subquery identifies flights that use airplanes with a seating capacity above 300.

```
-- Show all flights that use an airplane model with a capacity greater than 300
SELECT Flight_ID, Origin, Destination
FROM Flight
WHERE Airplane_ID IN (
    SELECT Airplane_ID
    FROM Airplane
    WHERE Capacity > 300
);
```

Output:

Flight_ID	Origin	Destination
3	Doha	London
5	Istanbul	Frankfurt
NULL	NULL	NULL

5) View

Airplane capacity view

This view displays airplane numbers, models, and capacities for airplanes with capacity greater than 200.

```
-- Create a View to display airplane numbers, models, and capacities greater than 200
CREATE VIEW Airplane_Capacity_View AS
SELECT Number AS Airplane_Number, Model, Capacity
FROM Airplane
WHERE Capacity > 200;
SELECT * FROM Airplane_Capacity_View;
```

Output:

Airplane_Number	Model	Capacity
SA200	A330	250
FN300	B777	300
QA400	A350	320
EM500	B787	250
EM700	A380	500
TK800	B747	410

2.6 Wasan's Queries

1) Basic SELECT

Airbus models details

This query retrieves all aircraft details from the Airplane table specifically for the 'A320' and 'A380' models.

```
-- Wasan's QUERIES
-- Retrieve all aircraft details for specific Airbus models (A320 and A380)
SELECT * FROM Airplane
WHERE Model IN ('A320', 'A380');
-- List the names and birth dates of passengers born before the year 1990
```

Output:

Airplane_ID	Model	Number	Capacity	Airline_ID
1	A320	SA100	180	1
7	A380	EM700	500	4
NULL	NULL	NULL	NULL	NULL

Passengers born before 1990

This query lists the first name, last name, and date of birth of all passengers who were born before January 1, 1990.

```
-- List the names and birth dates of passengers born before the year 1990
SELECT First_name, Last_name, DOB
FROM Passenger
WHERE DOB < '1990-01-01';
```

Output:

First_name	Last_name	DOB
Fahad	Harbi	1988-11-03
John	Smith	1985-07-09

2) Aggregate Functions

Total seating capacity per airline

This query calculates the total number of seats available for each airline provider by summing the capacity of all their airplanes.

```sql
-- Calculate the total seating capacity available for each airline provider
SELECT Airline_ID, SUM(Capacity) AS Total_Seats
FROM Airplane
GROUP BY Airline_ID;
```

Output:

Airline_ID	Total_Seats
1	430
2	300
3	320
4	750
5	160
6	410
7	190
8	200

Passenger distribution by nationality

This query counts the number of passengers belonging to each nationality to show the distribution of travelers across the system.

```sql
-- Group passengers by their nationality to see the distribution of travelers
SELECT Nationality, COUNT(Passenger_ID) AS Passenger_Count
FROM Passenger
GROUP BY Nationality;
```

Output:

Nationality	Passenger_Count
Saudi Arabia	3
Egypt	2
UAE	1
UK	1
Jordan	1
Qatar	1
USA	1

3) Join Between Tables

Bookings linked to airline names

This query joins the Booking, Flight, and Airline tables to identify which airline company is operating each specific ticket.

```
-- Link bookings to their respective airlines to identify which company is operating each ticket
SELECT B.Booking_ID, A.Name AS Airline_Name
FROM Booking B
JOIN Flight F ON B.Flight_ID = F.Flight_ID
JOIN Airline A ON F.Airline_ID = A.Airline_ID;
```

Output:

Result Grid | Filter Rows:

Booking_ID	Airline_Name
1	Saudi Airlines
2	Saudi Airlines
3	Saudi Airlines
4	Qatar Airways
5	Emirates
6	Turkish Airlines
7	Flynas
8	Saudi Airlines
9	British Airways
10	EgyptAir

Passenger names with flight destinations

This query links the Passenger, Booking, and Flight tables to retrieve the names of passengers alongside their final flight destinations.

```
-- Retrieve passenger names alongside their final flight destinations
SELECT P.First_name, P.Last_name, F.Destination
FROM Passenger P
JOIN Booking B ON P.Passenger_ID = B.Passenger_ID
JOIN Flight F ON B.Flight_ID = F.Flight_ID;
```

Output:

Result Grid | Filter Rows:

First_name	Last_name	Destination
Ali	Saleh	Jeddah
Sara	Ahmed	Jeddah
Fahad	Harbi	Dubai
Maryam	Saad	London
John	Smith	Istanbul
Lina	Hassan	Frankfurt
Omar	Farouq	Cairo
Nora	Ali	Doha
Huda	Salem	Dubai
James	Brown	Jeddah

4) Subqueries to Answer Specific Questions

Flights operated by Saudi Airlines

This subquery dynamically fetches the Airline_ID for 'Saudi Airlines' to filter and display all flight records operated by that specific company

```
-- Filter flights to show only those operated by 'Saudi Airlines' by fetching the ID dynamically
SELECT * FROM Flight
WHERE Airline_ID = (SELECT Airline_ID FROM Airline WHERE Name = 'Saudi Airlines');
```

Output:

Flight_ID	Origin	Destination	Departure_Time	Arrival_Time	Airline_ID	Airplane_ID
1	Riyadh	Jeddah	2025-01-01 08:00:00	2025-01-01 10:00:00	1	1
2	Jeddah	Dubai	2025-01-02 12:00:00	2025-01-02 15:00:00	1	2
7	Jeddah	Doha	2025-01-07 11:00:00	2025-01-07 13:00:00	1	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Passengers flying from Riyadh

This nested subquery locates the names of passengers who have booked any flight originating from the city of 'Riyadh'.

```
-- Locate passengers who have booked any flight originating from 'Riyadh' using nested IDs
SELECT First_name, Last_name FROM Passenger
WHERE Passenger_ID IN (
    SELECT Passenger_ID FROM Booking WHERE Flight_ID IN (
        SELECT Flight_ID FROM Flight WHERE Origin = 'Riyadh'
    )
);
```

Output:

First_name	Last_name
Ali	Saleh
Sara	Ahmed
Omar	Farouq

5) View

High-capacity fleet report

This view creates a dedicated report displaying the models, numbers, and capacities for all aircraft that have more than 250 seats, serving as a quick reference for largescale fleet operations.

```
);

-- ADDITIONAL VIEW: High-capacity aircraft report (Greater than 250 seats)
CREATE VIEW High_Capacity_Fleet AS
SELECT Model, Number, Capacity
FROM Airplane
WHERE Capacity > 250;

-- Display the results of the high-capacity fleet view
SELECT * FROM High_Capacity_Fleet;
```

Output:

Result Grid | Filter Rows:

Model	Number	Capacity
B777	FN300	300
A350	QA400	320
A380	EM700	500
B747	TK800	410

2.7 Retal's Queries

1) Basic SELECT Queries

Passengers from Egypt

This query retrieves the first name, last name, and phone number of passengers whose nationality is Egypt.

```
-- Show passenger names and phone numbers for passengers from Egypt
SELECT First_name, Last_name, Phone_Number
FROM Passenger
WHERE Nationality = 'Egypt';
```

Output:

	First_name	Last_name	Phone_Number
▶	Sara	Ahmed	0102233445
	Omar	Farouq	0124455667

Flights departing from Riyadh

This query displays flight ID, destination, and departure time for flights that originate from Riyadh.

```
-- Show flight details departing from Riyadh
SELECT Flight_ID, Destination, Departure_Time
FROM Flight
WHERE Origin = 'Riyadh';
```

Output:

	Flight_ID	Destination	Departure_Time
▶	1	Jeddah	2025-01-01 08:00:00
	6	Cairo	2025-01-06 07:00:00
*	NULL	NULL	NULL

2) Aggregate Functions

Number of passengers per nationality

This query groups passengers by nationality and counts the total number of passengers in each group.

```
-- Count number of passengers per nationality
SELECT Nationality, COUNT(Passenger_ID) AS Total_Passengers
FROM Passenger
GROUP BY Nationality;
```

Output:

	Nationality	Total_Passengers
▶	Saudi Arabia	3
	Egypt	2
	UAE	1
	UK	1
	Jordan	1
	Qatar	1
	USA	1

Average airplane capacity

This query calculates the average seating capacity of all airplanes in the database.

```
-- Calculate the average airplane capacity
SELECT AVG(Capacity) AS Average_Capacity
FROM Airplane;
```

Output:

	Average_Capacity
▶	276.0000

3) Join Between Tables

Booking details with passenger names

This query joins the Booking and Passenger tables to show booking information along with passenger names.

```
-- Show booking details with passenger full name
SELECT B.Booking_ID, P.First_name, P.Last_name, B.BookingDate
FROM Booking B
INNER JOIN Passenger P
    ON B.Passenger_ID = P.Passenger_ID;
```

Output:

Booking_ID	First_name	Last_name	BookingDate
1	Ali	Saleh	2025-01-01 07:00:00
2	Sara	Ahmed	2025-01-01 07:30:00
3	Fahad	Harbi	2025-01-02 11:00:00
4	Maryam	Saad	2025-01-03 03:00:00
5	John	Smith	2025-01-04 08:00:00
6	Lina	Hassan	2025-01-05 10:00:00
7	Omar	Farouq	2025-01-06 05:30:00
8	Nora	Ali	2025-01-07 09:40:00
9	Huda	Salem	2025-01-08 20:00:00
10	James	Brown	2025-01-09 05:15:00

Flights with airline names

This query joins the Flight and Airline tables to display flight details and their corresponding airline names.

```
-- Show flight origin, destination, and airline name
SELECT F.Flight_ID, F.Origin, F.Destination, A.Name AS Airline_Name
FROM Flight F
JOIN Airline A
    ON F.Airline_ID = A.Airline_ID;
```

Output:

Flight_ID	Origin	Destination	Airline_Name
1	Riyadh	Jeddah	Saudi Airlines
2	Jeddah	Dubai	Saudi Airlines
3	Doha	London	Qatar Airways
4	Dubai	Istanbul	Emirates
5	Istanbul	Frankfurt	Turkish Airlines
6	Riyadh	Cairo	Flynas
7	Jeddah	Doha	Saudi Airlines
8	London	Dubai	British Airways
9	Cairo	Jeddah	EgyptAir
10	Jeddah	Muscat	Oman Air

4) Subqueries to Answer Specific Questions

Passengers who booked flights to Dubai

This subquery identifies passengers who have booked flights with Dubai as the destination.

```
-- Show passengers who booked flights to Dubai
SELECT First_name, Last_name
FROM Passenger
WHERE Passenger_ID IN (
    SELECT Passenger_ID
    FROM Booking
    WHERE Flight_ID IN (
        SELECT Flight_ID
        FROM Flight
        WHERE Destination = 'Dubai'
    )
);
```

Output:

	First_name	Last_name
▶	Fahad	Harbi
	Huda	Salem

Flights using airplanes with capacity less than 200 This subquery retrieves flights that use airplanes with a seating capacity below 200.

```
-- Show flights that use airplanes with capacity less than 200
SELECT Flight_ID, Origin, Destination
FROM Flight
WHERE Airplane_ID IN (
    SELECT Airplane_ID
    FROM Airplane
    WHERE Capacity < 200
);
```

Output:

	Flight_ID	Origin	Destination
▶	1	Riyadh	Jeddah
	7	Jeddah	Doha
	9	Cairo	Jeddah
●	NULL	NULL	NULL

5) View

Booking report view

This view creates a virtual table that combines booking, passenger, and flight information for reporting purposes.

```
-- Create a view showing booking info with passenger and flight destination
CREATE VIEW Booking_Report AS
SELECT B.Booking_ID,
       P.First_name,
       P.Last_name,
       F.Origin,
       F.Destination,
       B.TicketClass
FROM Booking B
JOIN Passenger P ON B.Passenger_ID = P.Passenger_ID
JOIN Flight F ON B.Flight_ID = F.Flight_ID;

SELECT * FROM Booking_Report;
```

Output:

Booking_ID	First_name	Last_name	Origin	Destination	TicketClass
1	Ali	Saleh	Riyadh	Jeddah	Economy
2	Sara	Ahmed	Riyadh	Jeddah	Business
3	Fahad	Harbi	Jeddah	Dubai	Economy
4	Maryam	Saad	Doha	London	First
5	John	Smith	Dubai	Istanbul	Economy
6	Lina	Hassan	Istanbul	Frankfurt	Business
7	Omar	Farouq	Riyadh	Cairo	Economy
8	Nora	Ali	Jeddah	Doha	Business
9	Huda	Salem	London	Dubai	Economy
10	James	Brown	Cairo	Jeddah	First